

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

Criteria	Understanding of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total(10)
Weightage	25%	30%	20%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I Vaishnavee BV(192571071) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Age of child	Photograph:		
	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID Player Name Points Scored

P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17

Player ID Player Name Points Scored

P13	Player M	19
P14	Player N	21
P15	Player O	23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set

young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

- (a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
- (b) Draw the boxplots for *age* and *%fat*.
- (c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

1. χ^2 test

observed frequencies (O)

- 5-6 yrs: A=18, B=22, C=20 (Total=60)
- 7-8 yrs: A=2, B=28, C=40 (Total=70)
- 9-10 yrs: A=20, B=10, C=40 (Total=70)
- Column total: A=40, B=60, C=100

Expected Frequency ($E = \frac{\text{Row total} \times \text{Col total}}{\text{grand total}}$):

- 5-6 yrs: $E_A = 12, E_B = 18, E_C = 30$
- 7-8 yrs: $E_A = 14, E_B = 21, E_C = 35$
- 9-10 yrs: $E_A = 14, E_B = 21, E_C = 35$

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 = \frac{6^2}{12} + \frac{4^2}{18} + \frac{(-10)^2}{14} + \frac{7^2}{21} + \frac{5^2}{35} + \frac{6^2}{14}$$

$$+ \frac{(-11)^2}{12} + \frac{5^2}{35}$$

$$\chi^2 = 3 + 0.889 + 3.333 + 10.286 + 2.33 + 0.7$$

$$+ 2.57 + 5.762 + 0.714 = 29.60$$

• Degrees of freedom

$$df = (3-1) \times (3-1) = 4$$

critical value at $\alpha = 0.05$ is 9.488

Since $29.60 > 9.488$ there is a statistically significant relationship b/w age and photograph preference

• what's wrong with the study?
confounding variables, sample bias

• R implementation
data ← data.frame

$$A = c(18, 2, 20)$$

$$B = c(22, 28, 10)$$

$$C = c(20, 40, 40)$$

$$\text{cov_BC} \leftarrow \text{cov}(data\$B, data\$C)$$

$$\text{cov_BC} \leftarrow \text{cov}(data\$B, data\$C)$$

2. Outlier detection & Box Plot analysis

Data

Points scored $n=15$

Quantile calculation:

- Median (Q_2): 16 (8th value)
- First quartile (Q_1): 12
- Third quartile (Q_3): 20
- Interquartile Range (IQR): $Q_3 - Q_1$
 $= 20 - 12 = 8$

- Outlier Boundaries

$$\text{Lower Bound: } Q_1 - 1.5 \times IQR$$

$$= 12 - 1.5(8) = 0$$

$$\text{Upper Bound: } Q_3 + 1.5 \times IQR$$

$$= 20 + 1.5(8) = 32$$

Q3

- Inface Player J (P_{10}) scored 35 points - outlier

$$E(S) = - \left(\frac{9}{15} \log_2 \frac{9}{15} + \frac{6}{15} \log_2 \frac{6}{15} \right)$$

$$\approx 0.971$$

$$E(S|\text{employment})$$

$$= \frac{6}{15} (0) + \frac{4}{15} (0.811) + \frac{5}{15} (1)$$

$$= 0.216$$

Q4

Mean	65.78	62.11	class A
median	66	60	class A
range	60	34	class A

class A $\rightarrow$ higher overall performance
 higher variance & spread

class B $\rightarrow$ lower overall score but
 greater consistency in performing.

5. Age data

$n=18$: 23, 23, 27, 27, 39, 41, 47, 49,
50, 52, 54, 54, 56, 58, 58,
60, 61

$$\text{mean} = \frac{836}{18} = 46.44$$

median: $\frac{18}{2}$ Avg of 9th & 10th

$$\frac{50+52}{2} = 51.00$$

standard deviation (s)
13.22

% fat data ($n=18$)

7.8, 9.5, 17.8, 25.9, 26.5, 27.2,
27.4, 28.8, 30.2, 31.2, 34.4, 37.9,
33.4, 34.1, 34.6, 35.7, 41.2,
42.5

$$\text{mean} = \frac{498.1}{18} = 27.67$$

$$\text{median} \rightarrow \frac{30.2 + 31.2}{2} = 30.7$$

standard deviation (s) = 9.21


```
import matplotlib.pyplot as plt
import scipy.stats as stats
```

```
age = [23, 23, 27, 27, 39, 41, 43, 49, 50,
        52, 54, 54, 56, 57, 58, 58, 60, 60, 61]
```

```
fat = [9.5, 26.5, 7.8, 31.4, 25.9, 27.4,
        27.2, 31.2, 34.6, 42.5, 28.8, 33.4,
        30.2, 34.1, 32.9, 41.2, 35.7]
```

```
plt.figure(figsize=(8,4))
plt.boxplot([age, fat], labels=['Age', 'fat'])
plt.title('Boxplot of Age & f. fat')
plt.show()
```

```
plt.scatter(age, fat)
plt.xlabel('Age')
plt.ylabel('f. fat')
plt.title('Scatter plot: Age vs. f. fat')
plt.show()
```

```
stats.probplot(age, dist="norm",
               plot=plt)
plt.show()
```

```
stats.probplot(fat, dist="norm",
               plot=plt)
plt.title('Q-Q plot: Age vs. fat')
plt.show()
```